

# TSA Meeting — Slide 4 Study Note

## SAMBP Framework: Three-Tier Architecture

Architecture Overview · EKF Oracle · WAPC · Cybersecurity Layer

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### Slide Overview and Narrative

#### What This Slide Shows

Slide 4 is the **architectural map** of the entire thesis. Before diving into any individual algorithm, the TSA needs to see how all the pieces fit together. The slide answers: *what is SAMBP, in one picture?*

The architecture flows left to right across three tiers:

1. **Tier 1 — Parameter Oracle** (EKF Digital Twin): A single, continuously-running Extended Kalman Filter that tracks the three latent network state variables ( $\hat{k}_{ibr}$ ,  $\hat{Z}_{line}$ ,  $\hat{Y}_{ibr}$ ) in real time. Every protection module in Tier 2 queries this oracle for the current  $\hat{k}_{ibr}$  value before computing any threshold.
2. **Tier 2 — Protection Modules** (six algorithms): The six protection elements are arranged in three pairs, each pair representing a primary-to-secondary dependency:
  - SyncOC (inverse curve) → Microgrid (mode-switch)
  - 87L (adaptive slope) → Dist. 21 (zone adapt.)
  - 87T/87B (SPRT) → PINN (physics-informed classifier)
3. **Tier 3 — WAPC** (topology-aware coordination): The Wide-Area Protection Coordinator integrates the output of all Tier 2 modules, performs PMU-based fault localisation, verifies N-1 security, and dispatches coordinated setting updates via IEC 61850 GOOSE.

**Cutting across all tiers:** A Cybersecurity Layer applies HMAC-SHA256 authentication to GOOSE messages and adversarial hardening to the PINN classifier, protecting the entire communication fabric against spoofing, replay, and adversarial injection attacks.

**The key principle stated on the slide:** *“Each module queries the EKF oracle for  $\hat{k}_{ibr}$ , computes adaptive thresholds via the bounded update law, and executes a confidence-gated (Stage-2) veto before tripping.”* This one sentence describes the common three-step action pattern of every protection element in the framework.

## Architecture at a Glance

| Tier | Name                   | Chapter(s)          | Primary Function                                                              |
|------|------------------------|---------------------|-------------------------------------------------------------------------------|
| 1    | Parameter Oracle (EKF) | Ch 6 (Digital Twin) | Real-time ( $\hat{k}_{ibr}$ , $\hat{Z}_{line}$ , $\hat{Y}_{ibr}$ ) estimation |
| 2a   | SyncOC                 | Ch 3                | Adaptive generator over-current, IEC inv. curve                               |
| 2b   | 87L line diff          | Ch 3                | Adaptive bias slope $k_m^*(k_{ibr})$                                          |
| 2c   | 87T / 87B              | Ch 6 / Ch 6         | SPRT inrush discriminator / bus differential                                  |
| 2d   | Microgrid              | Ch 5                | Mode-switch protection; ROCOF islanding                                       |
| 2e   | Dist. 21               | Ch 5                | Quadrilateral Zone 1 reach correction                                         |
| 2f   | PINN                   | Ch 6                | Physics-informed ML fault classifier                                          |
| 3    | WAPC                   | Ch 7                | Topology-aware coordination; N-1 security                                     |
| ⊥    | Cybersecurity          | Ch 8                | HMAC GOOSE; adversarial PINN hardening                                        |

## Engineering Motivation: Why Three Tiers?

## The Three-Tier Rationale

A single monolithic adaptive relay would create three unacceptable problems:

1. **Single point of failure:** If the estimator fails (e.g., under a cyberattack that injects false measurements), all protection fails simultaneously.
2. **Inconsistent  $\hat{k}_{ibr}$ :** If each relay runs its own LM estimator independently, they may produce different  $\hat{k}_{ibr}$  values, causing *coordination failures* where the primary relay sees  $k_{ibr} = 0.3$  (fast trip) and the backup sees  $k_{ibr} = 0.5$  (slow trip), resulting in CTI violation.
3. **No wide-area view:** A local relay cannot verify N-1 security or dispatch coordinated settings fleet-wide.

The three-tier architecture separates these concerns:

- **Tier 1** provides one consistent  $\hat{k}_{ibr}$  to all modules (single source of truth).
- **Tier 2** runs fast local protection actions (20–80 ms trip decisions) without waiting for Tier 3.
- **Tier 3** runs slow wide-area coordination (>100 ms) without blocking Tier 2.

The cybersecurity layer is *orthogonal* — it secures communication at every tier transition without adding algorithmic latency to Tier 2.

## Tier 1: Parameter Oracle — The EKF Digital Twin

### EKF State and Measurement Models

The EKF digital twin maintains a continuous estimate of three network state variables:

$$\mathbf{x}_k = \begin{bmatrix} k_{\text{ibr}} \\ Z_{\text{line}} \\ Y_{\text{ibr}} \end{bmatrix}_k \quad (1)$$

The state transition model (between faults):

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, Q) \quad (2)$$

where  $Q = \text{diag}(\sigma_{k_{\text{ibr}}}^2, \sigma_{Z}^2, \sigma_Y^2)$  represents the expected drift rate of each state variable.

The measurement model (from IED phasors every 20 ms):

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, R) \quad (3)$$

where  $h(\cdot)$  is the power flow model mapping  $(k_{\text{ibr}}, Z_{\text{line}}, Y_{\text{ibr}})$  to observable bus voltages and branch currents.

### What the Oracle Provides to Each Tier-2 Module

Every 20 ms (one power cycle), the EKF outputs:

- $\hat{k}_{\text{ibr}}$  with 95% confidence interval (CI width  $< 0.05$  pu).
- $\hat{Z}_{\text{line}}$  (used by Dist. 21 for Zone 1 correction).
- $\hat{Y}_{\text{ibr}}$  (used by 87L bias computation).
- $\text{RMSE} \leq 0.024$  pu for  $\hat{k}_{\text{ibr}}$  (validated in TR-09).

**Non-blocking design:** The EKF runs as an asynchronous background process. If the EKF result is not yet available when a Tier-2 module needs  $\hat{k}_{\text{ibr}}$  (e.g., fault occurs during EKF prediction step), the module uses the last valid estimate with the confidence score degraded accordingly. This ensures protection continuity even during EKF computation cycles.

### EKF vs. LM: When Each Is Used

| Method       | When Used                | Update Rate     | Role                                  |
|--------------|--------------------------|-----------------|---------------------------------------|
| EKF (Tier 1) | Continuous (inter-fault) | 20 ms / cycle   | Tracks $k_{\text{ibr}}$ drift slowly  |
| LM (Tier 2)  | During fault window      | 3–5 ms / window | Identifies fault parameters precisely |

The EKF provides the *initial condition* and *pre-fault estimate*; the LM refines it *post-fault*. Formally, the EKF prior  $p(\mathbf{x}|\mathbf{y}_{1:k-1})$  seeds the LM starting point  $\boldsymbol{\theta}^{(0)}$ , reducing LM iterations from 20–30 (cold start) to 10–15 (warm start).

## Tier 2: Protection Modules — The Six Algorithms

### The Three-Step Action Pattern (Key Principle on Slide)

Every Tier-2 module follows the same three-step action:

1. **Query:** Read  $\hat{k}_{ibr}$  from the EKF oracle.
2. **Compute:** Derive the adaptive threshold using the bounded update law (e.g.,  $k_m^*(\hat{k}_{ibr})$  for 87L,  $I_{set}(\hat{k}_{ibr})$  for SyncOC,  $X_{reach,corr}$  for Dist. 21).
3. **Gate:** Check  $\gamma \geq \gamma_{th}$  (Stage-2 confidence veto) before authorising a trip output.

This pattern is the same across all six modules; only the threshold formula in Step 2 differs.

### Module-Pair Dependencies

The three module pairs on the slide reflect operational couplings:

**SyncOC → Microgrid:** The generator OC relay must be aware of the microgrid's operating mode (grid-connected vs. islanded), because the fault current magnitude is different in each mode. In islanded mode, the only fault current source is the local generation; in grid-connected mode, the grid contributes. The SyncOC setting group switches automatically when the Microgrid module broadcasts a mode-switch GOOSE.

**87L → Dist. 21:** The line differential and distance relays protect the same line from opposite ends. The 87L element has primary responsibility; Dist. 21 is backup. The Dist. 21 Zone 1 reach correction uses the same  $\hat{k}_{ibr}$  as 87L, ensuring they are both calibrated to the same IBR state.

**87T/87B → PINN:** The transformer and bus differential elements generate event records (trip signals, phasor snapshots) that feed the PINN classifier. The PINN provides a secondary check: if 87T trips on inrush (false trip), the PINN post-classifies the event and can issue a trip-retract command to the breaker if the breaker has not yet fully opened (within <10 ms).

### Confidence Gate: Why It Eliminates False Trips Under CT Saturation

The slide's observation: *“Confidence gate  $\gamma$  eliminates false trips under CT saturation or communication latency.”*

Under CT saturation, the measured current  $\tilde{I}$  deviates from the true  $I$  by a nonlinear error. The LM residual  $\|\mathbf{r}\|^2$  increases because the physical model  $\mathbf{x}(t; \boldsymbol{\theta})$  can no longer fit the saturated waveform. This increases  $(1 - R^2)$ , which decreases  $\gamma_R$ , which reduces  $\gamma$  below  $\gamma_{th}$ . The Stage-2 gate *blocks the trip* until the estimate is trustworthy.

Under communication latency, the EKF estimate  $\hat{k}_{ibr}$  at the Tier-2 module may be stale. The EKF covariance  $P_k$  grows as the latency increases (process noise accumulation). The confidence score component  $\gamma_J$  (Jacobian conditioning) degrades when  $P_k$  is large, pulling  $\gamma$  below  $\gamma_{th}$  and blocking the trip until fresh data arrives.

### Quantified Gate Performance

From the 110-scenario validation (TR-08):

| Scenario              | False trips (no gate) | False trips (with gate) | Reduction   |
|-----------------------|-----------------------|-------------------------|-------------|
| CT saturation 5%      | 16                    | 0                       | −100%       |
| CT saturation 3%      | 8                     | 0                       | −100%       |
| Comm. latency 20 ms   | 4                     | 0                       | −100%       |
| High-resistance fault | 3                     | 1                       | −67%        |
| <b>Total</b>          | <b>31</b>             | <b>1</b>                | <b>−97%</b> |

The one remaining false trip (high-resistance fault,  $R_F > 50 \Omega$ ) is a genuine edge case where  $I_{\text{diff}}$  is ambiguous; it is flagged as a “supervised review” case and does not auto-trip.

## Tier 3: WAPC — Topology-Aware Coordination

### Four WAPC Capabilities

- C15 — PMU Fault Localisation:** Voltage-dip triangulation identifies the faulted line by maximising  $\Delta V_i + \Delta V_j$  over all line candidates  $(i, j)$ . Uses Dijkstra electrical distance for the dip model. Correct in  $<10$  ms at 50 PMU frames/s.
- C16 — CTI-Graded Settings Dispatch:** After localising the fault, the WAPC re-optimises relay settings for the post-fault topology using projected-gradient coordination subject to  $t_{\text{primary}} + \text{CTI} \leq t_{\text{backup}}$  ( $\text{CTI} \geq 0.3$  s per IEC 60255-151). New settings dispatched via MMS in  $<100$  ms.
- C17 — HMAC-SHA256 GOOSE Integrity:** Every GOOSE message is authenticated with a 32-byte HMAC tag using IEEE 1588 PTP timestamp and sequence number. Spoofing success rate:  $12.3\% \rightarrow <0.001\%$ . Latency overhead:  $0.72$  ms ( $\ll 4$  ms IEC 61850 P2 limit).
- C18 — Closed-Loop Adaptation Protocol:** CUSUM drift detection  $\rightarrow$  LM re-estimation  $\rightarrow$  settings dispatch  $\rightarrow$  GOOSE re-authentication. Full cycle in  $<200$  ms, maintaining continuous protection during network state changes.

### N-1 Security Verification

Before dispatching any setting change, the WAPC verifies N-1 security: it checks that *all* lines in the post-fault topology have valid protection coverage, even if one additional element trips. If N-1 security is violated, the WAPC halts setting dispatch and raises an alarm for human review rather than applying a setting that could leave the network unprotected.

## Cybersecurity Layer: HMAC and Adversarial PINN Hardening

### Why Adaptive Protection Introduces a New Attack Surface

Fixed relay settings are offline configurations — an attacker cannot change a trip threshold by manipulating live data. Adaptive protection creates an attack surface: if an attacker can inject false current measurements, they can manipulate  $\hat{k}_{ibr}$  and thereby cause a misconfigured relay to miss a genuine fault (security failure) or trip on normal load (reliability failure). The cybersecurity layer addresses both attack vectors.

### HMAC-SHA256 for GOOSE Integrity

The GOOSE authentication tag:

$$\text{tag}_t = \text{HMAC}_k(\text{DataSet}_t \parallel t_{\text{UTC}} \parallel \text{SqNum}) \quad (4)$$

where  $k$  is a 256-bit shared key,  $t_{\text{UTC}}$  is the microsecond-precision IEEE 1588 PTP times-tamp, and SqNum is the GOOSE sequence number. The receiver rejects any message where:

- The HMAC tag does not verify (spoofing detection), or
- $|t_{\text{UTC}} - t_{\text{now}}| > \delta_{\text{replay}} = 5 \text{ ms}$  (replay detection).

### Adversarial PINN Hardening

A CT-injection attack perturbs the input by  $\delta$  with  $\|\delta\|_{\infty} \leq \varepsilon$  to cause the PINN to misclassify. The physics constraint in the PINN loss:

$$\mathcal{L}_{\text{PINN}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{phys}} \|\mathbf{KVL}(\mathbf{x})\|^2 + \lambda_{\text{phys}} \|\mathbf{KCL}(\mathbf{x})\|^2 \quad (5)$$

forces any successful adversarial perturbation to also satisfy KVL/KCL. This raises the minimum perturbation energy  $\varepsilon^*$  required for misclassification, making the attack substantially harder than against an unconstrained neural network.

### Information Flow: End-to-End Data Path

#### Tracing One Fault Event Through All Three Tiers

1. **Pre-fault (continuous):** EKF receives IED phasor data every 20 ms; outputs  $(\hat{k}_{ibr}, \hat{Z}_{line}, \hat{Y}_{ibr})$  to all Tier-2 modules.
2. **Fault detection ( $t = 0$ ):** 87L detects  $I_{\text{diff}} > k_m I_{\text{bias}}$ . Queries EKF for  $\hat{k}_{ibr}$ ; computes  $k_m^*(\hat{k}_{ibr})$ .
3. **Stage-2 veto ( $t = 0 \rightarrow 10 \text{ ms}$ ):** Runs LM on first 10 ms of fault window; computes  $\gamma$ . If  $\gamma \geq 0.85$ : issues GOOSE trip. If  $\gamma < 0.85$ : waits one more cycle.
4. **HMAC sign ( $t \approx 10 \text{ ms}$ ):** GOOSE trip message is tagged with HMAC and sent to circuit breaker IED.
5. **WAPC receives ( $t \approx 15 \text{ ms}$ ):** Fault localisation ( $\hat{L}$ ) by voltage-dip triangulation. N-1 check.
6. **Settings re-dispatch ( $t \approx 100 \text{ ms}$ ):** WAPC computes post-fault topology

settings; dispatches via MMS.

7. **HMAC re-key** ( $t \approx 200$  ms): Closed-loop protocol completes; CUSUM resets; EKF re-initialises for new topology.

## Cross-Thesis Connections

| Chapter          | What It Contributes to the Architecture                                | Where It Appears on Slide              |
|------------------|------------------------------------------------------------------------|----------------------------------------|
| Ch 3 (87L)       | Adaptive slope $k_m^*(k_{ibr})$ ; Stage-2 gate at $\gamma_{th} = 0.85$ | Middle tier: “87L / Adaptive slope”    |
| Ch 4 (SyncOC)    | LM fault current estimation; CTI coordination                          | Middle tier: “SyncOC / Inv. curve”     |
| Ch 5 (87T/87B)   | SPRT inrush discriminator; bus diff                                    | Middle tier: “87T / 87B / SPRT”        |
| Ch 5 (Dist. 21)  | Quadrilateral Zone 1 correction                                        | Middle tier: “Dist. 21 / Zone adapt.”  |
| Ch 5 (Microgrid) | Mode-switch protection; ROCOF                                          | Middle tier: “Microgrid / Mode-switch” |
| Ch 6 (DT/EKF)    | Non-blocking EKF oracle; $RMSE \leq 0.024$ pu                          | Tier 1: “Parameter Oracle”             |
| Ch 6 (PINN)      | Physics-constrained fault classifier                                   | Middle tier: “PINN”                    |
| Ch 7 (WAPC)      | PMU localisation; CTI dispatch; C15–C18                                | Tier 3: “WAPC”                         |
| Ch 8 (Security)  | HMAC GOOSE; adversarial hardening                                      | Cybersecurity Layer                    |

## Key Equations Reference Card

## All Equations You Must Know for This Slide

**EKF state:**  $\mathbf{x}_k = [k_{\text{ibr}}, Z_{\text{line}}, Y_{\text{ibr}}]^\top$  (E1)

**EKF process:**  $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{w}_k, \quad Q = \text{diag}(\sigma_{k_{\text{ibr}}}^2, \sigma_Z^2, \sigma_Y^2)$  (E2)

**Three-step action:** Query  $\rightarrow$  Compute  $k_m^*(\hat{k}_{\text{ibr}}) \rightarrow$  Gate  $\gamma \geq \gamma_{\text{th}}$  (E3)

**WAPC fault localise:**  $\hat{L} = \arg \max_{L_{ij}} [\Delta V_i + \Delta V_j]$  (E4)

**CTI constraint:**  $t_{\text{primary}} + \text{CTI} \leq t_{\text{backup}}, \quad \text{CTI} \geq 0.3 \text{ s}$  (E5)

**HMAC tag:**  $\text{tag}_t = \text{HMAC}_k(\text{DataSet}_t \parallel t_{\text{UTC}} \parallel \text{SqNum})$  (E6)

**PINN loss:**  $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{phys}}(\|\mathbf{KVL}\|^2 + \|\mathbf{KCL}\|^2)$  (E7)

**Replay reject:**  $|t_{\text{UTC}} - t_{\text{now}}| > 5 \text{ ms} \Rightarrow \text{discard}$  (E8)

**HMAC risk reduction:** Spoofing:  $12.3\% \rightarrow < 0.001\%$ , overhead  $0.72 \text{ ms}$  (E9)

## Timed Speaker Script (3 Minutes)

## What to Say at the TSA Meeting

**[0:00–0:35] The Big Picture**

“This slide shows the three-tier SAMBP architecture: how every algorithm in the thesis connects into a single system. The key design principle is separation of concerns. Tier 1 is the Parameter Oracle: a continuously-running Extended Kalman Filter that tracks the IBR penetration ratio  $\hat{k}_{\text{ibr}}$ , line impedance, and IBR admittance from IED phasor data every 20 milliseconds. All six protection modules in Tier 2 query this oracle before every threshold computation. This gives the system a single, consistent  $\hat{k}_{\text{ibr}}$  estimate, which is critical for relay coordination.”

**[0:35–1:15] Tier 2 and the Three-Step Pattern**

“The six Tier-2 modules are the protection algorithms. Every one of them follows the same three-step action. First, query the EKF oracle for  $\hat{k}_{\text{ibr}}$ . Second, compute the adaptive threshold — whether that is the adaptive bias slope for 87L, the adaptive pickup for SyncOC, or the corrected Zone 1 reach for the distance relay. Third, execute the Stage-2 confidence veto: only trip if  $\gamma \geq \gamma_{\text{th}}$ . This Stage-2 gate is the key safety mechanism: under CT saturation or communication latency, the confidence score drops below the threshold and the trip is withheld until the estimate is trustworthy. In the 110-scenario validation, this eliminated 97% of false trips without causing any missed trips.”

**[1:15–1:50] Tier 3: WAPC**



“Tier 3 is the Wide-Area Protection Coordinator. It adds four capabilities that no individual relay can provide. PMU-based fault localisation using voltage-dip triangulation. N-1 security verification to ensure the post-fault topology has full protection coverage. Automatic CTI-graded settings re-dispatch via IEC 61850 MMS within 100 milliseconds. And a closed-loop protocol that chains fault detection, re-estimation, settings update, and GOOSE re-authentication into one 200 millisecond cycle.”

#### [1:50–2:30] Cybersecurity Layer

“The cybersecurity layer cuts across all three tiers because adaptive protection introduces a new attack surface: if an adversary can corrupt the  $\hat{k}_{ibr}$  estimate, they can cause the relay to malfunction. Two mechanisms address this. First, HMAC-SHA256 authentication on every GOOSE message prevents spoofing and replay — spoofing success dropped from 12.3% to below 0.001% with only 0.72 millisecond overhead. Second, adversarial hardening of the PINN classifier: the physics constraints embedded in the loss function force any successful adversarial perturbation to satisfy KVL and KCL, substantially raising the attack energy required.”

#### [2:30–3:00] The Thesis Map

“Every chapter of the thesis corresponds to one box in this diagram. Chapter 1 is the introduction, Chapter 2 is the mathematical foundations that all modules share. Chapters 4 through 8 are the six Tier-2 protection algorithms. Chapter 6 is the EKF digital twin. Chapter 10 is the PINN. Chapter 7 is the WAPC. Chapter 8 is the cybersecurity layer. The architecture is the thesis.”

## Anticipated Q&A — Five Likely Committee Questions

**Q1: Is the EKF oracle a single point of failure? What happens if it crashes?**

**Answer:** Yes, the EKF oracle is a potential single point of failure, and this is a design limitation acknowledged in the thesis. Three mitigations are implemented: (a) The EKF is non-blocking — each Tier-2 module stores the last valid  $\hat{k}_{ibr}$  estimate with a timestamp and uses it for up to 500 ms before raising a “stale estimate” alarm. (b) Each module has a conservative fallback: if the EKF is unavailable, the module defaults to the commissioning-time  $k_{ibr}$  estimate (typically  $k_{ibr} = 0.3$  for a mixed-source feeder). (c) The WAPC includes an EKF health watchdog that flags the fault condition and alerts operators. Future work (Ch 13) includes a hot-standby EKF replica.

**Q2: The three module pairs seem arbitrary. Why SyncOC with Microgrid and not SyncOC with 87L?**

**Answer:** The pairings reflect operational coupling, not algorithmic similarity. SyncOC and Microgrid are paired because the generator OC relay must change its setting group when the microgrid transitions between grid-connected and islanded modes; the Microgrid module triggers the SyncOC mode-switch via GOOSE. 87L and Dist. 21 are paired because they protect the same physical line (primary and

backup); they share the same  $\hat{k}_{ibr}$  input for consistent threshold computation. 87T/87B and PINN are paired because the transformer and bus events feed directly into the PINN classifier for post-classification. The pairings are architectural dependencies, not groupings of similar algorithms.

**Q3: The HMAC key is shared at commissioning. What if an IED is compromised and the key is extracted?**

**Answer:** This is the key compromise vulnerability acknowledged in Chapter 12. The current implementation uses a static shared key, which is the standard practice in IEC 62351-6. Future work (listed in Chapter 13) includes automatic key rotation per IEC 62351-8, which rotates the HMAC key every 24 hours using a public-key infrastructure. Until key rotation is implemented, the mitigation is physical security of the IED hardware (tamper-evident enclosures) and network segmentation (VLAN isolation prevents an attacker from accessing the substation LAN even after key compromise of one IED).

**Q4: Can the confidence gate cause a missed trip if the CT saturation persists for multiple cycles?**

**Answer:** This is the fundamental security-vs-dependability trade-off in the Stage-2 gate. Under persistent CT saturation (lasting >80 ms), the gate would continue to block the trip, which is a missed trip. Two backup mechanisms address this: (a) After 80 ms (4 cycles) of blocked trips, the gate is overridden and the relay trips on the last known threshold — this is the “security timeout” in the stage-2 logic. (b) The WAPC’s PMU-based fault localisation does not rely on CT data; it uses bus voltages from PMUs, which are unaffected by CT saturation. The WAPC can issue an independent trip command within 15 ms via the IEC 61850 MMS channel.

**Q5: How does the PINN trip-retract command work within 10 ms? Is that feasible?**

**Answer:** The trip-retract applies only to specific relay types that have an “output hold” function: the relay asserts the trip contact for a configurable hold time (default 200 ms) before the contact opens and the breaker mechanism activates. Modern numerical relays (e.g., Siemens SIPROTEC 5) support an output hold of 10–50 ms. The PINN post-classification receives the 87T trip GOOSE via the substation LAN (<1 ms latency on a 100 Mbps VLAN), runs the PINN inference (<2 ms on an edge GPU), and sends a retract GOOSE command. Total: <3 ms, well within the 10 ms hold window. This mechanism was validated in software-in-the-loop (SIL) testing.

**Summary for the TSA Committee**

The three-tier SAMBP architecture separates parameter estimation (Tier 1 EKF), fast local protection (Tier 2 six modules) and wide-area coordination (Tier 3 WAPC) into independent layers with well-defined interfaces. The common three-step pattern (query → compute → gate) and the Stage-2 confidence veto are the architectural glue that connects Ch 2's mathematics to Chs 3–8's implementations. The cybersecurity layer protects the communication infrastructure that the adaptive system depends on.